

3.6 The Chain Rule

Motivation:

Suppose we want to find the first derivative of $f(x) = (2x+1)^3$. One tedious way to do this would be to multiply out the formula for the function, and then apply the power rule.

Specifically, we can note: $f(x) = 8x^3 + 12x^2 + 6x + 1$ which implies $f'(x) = 24x^2 + 24x + 6$.

VERY INTERESTINGLY, we notice that if we simplify this, we get:

$$f'(x) = 6(4x^2 + 4x + 1) = 6(2x+1)^2 = 3(2x+1)^2 \cdot 2.$$

So in looking for a shortcut to compute this derivative, it seems like we should differentiate the *outside* first, and then multiply that result by the derivative of what is in the *inside*. More formally, if we let $g(x) = x^3$ and $h(x) = 2x+1$, then $f(x) = (g \circ h)(x) = g(h(x))$.

Moreover, we have discovered that $f'(x) = g'(h(x))h'(x)$.

This idea leads to one of the most important differentiation rules: *The Chain Rule*.

Theorem (The Chain Rule): Suppose g is differentiable at x and f is differentiable at $g(x)$. Suppose that $F(x) = (f \circ g)(x) = f(g(x))$. Then, F is differentiable at x and:

$$F'(x) = f'(g(x)) \cdot g'(x)$$

In Leibniz notation, if $y = g(x)$, then we can think of f as a function of y and the chain rule becomes:

$$\frac{dF}{dx} = \frac{df}{dy} \frac{dy}{dx}$$

A partial proof of this theorem is presented in the textbook. A full proof can be found online.

Example 1: Find $F'(x)$ if $F(x) = \sqrt{x^2 + 1}$.

We first rewrite the function as $F(x) = (x^2 + 1)^{1/2}$.

Now we work our way from the outside to the inside and obtain:

$$F'(x) = \frac{1}{2}(x^2 + 1)^{-1/2} \cdot \frac{d}{dx}(x^2 + 1) = \frac{1}{2\sqrt{x^2 + 1}} \cdot 2x = \frac{x}{\sqrt{x^2 + 1}}$$

Note: Formally, in the above example (using the notation introduced in the statement of the chain rule) we have that $f(x) = \sqrt{x}$ and $g(x) = x^2 + 1$. $F(x) = (f \circ g)(x) = f(g(x))$. However, we usually intuitively think about differentiating from the outside to the inside when we use the chain rule in practice.

Example 2: Find the derivative of each of the following functions.

a) $f(x) = \sin(x^2)$

We compute: $f'(x) = \cos(x^2) \cdot \frac{d}{dx}(x^2) = \cos(x^2) \cdot 2x = 2x \cos(x^2)$

b) $g(x) = \sin^2(x)$

We first note that $g(x) = \sin^2(x) = [\sin(x)]^2$

We compute: $g'(x) = 2[\sin(x)]^1 \cdot \frac{d}{dx}[\sin(x)] = 2\sin(x)\cos(x)$

c) $h(x) = e^{3x}$

We compute: $h'(x) = e^{3x} \cdot \frac{d}{dx}(3x) = e^{3x} \cdot 3 = 3e^{3x}$

d) $r(x) = \frac{1}{(x^3 + 2)^5}$

We first note that $r(x) = \frac{1}{(x^3 + 2)^5} = (x^3 + 2)^{-5}$

We compute: $r'(x) = -5(x^3 + 2)^{-6} \cdot \frac{d}{dx}(x^3 + 2) = \frac{-5}{(x^3 + 2)^6} \cdot (3x^2) = \frac{-15x^2}{(x^3 + 2)^6}$

Note: This question could have also been done using the Quotient Rule, however it would be a slightly more complicated approach.

We will now examine some more complicated situations which involve the chain rule.

Example 3: Suppose that f , g , and h are differentiable functions. Find a formula for the first derivative of $F(x) = f(g(h(x)))$.

We apply the chain rule twice and discover that:

$$F'(x) = f'(g(h(x))) \cdot \frac{d}{dx}[g(h(x))] = f'(g(h(x))) \cdot g'(h(x)) \cdot \frac{d}{dx}[h(x)] = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x).$$

Note: This shows that we can also work our way from the outside to the inside when we are dealing with many compositions. Also it is clear that this pattern will continue to occur if we start with more than 3 functions.

Example 4: Answer each of the following:

a) Find $\frac{dy}{dx}$ if $y = e^{\sec(x^3)}$.

Again we work from the outside to the inside:

$$\begin{aligned} \frac{dy}{dx} &= e^{\sec(x^3)} \cdot \frac{d}{dx}[\sec(x^3)] = e^{\sec(x^3)} \cdot \sec(x^3) \tan(x^3) \cdot \frac{d}{dx}[x^3] = e^{\sec(x^3)} \cdot \sec(x^3) \tan(x^3) \cdot 3x^2 \\ &= 3x^2 \sec(x^3) \tan(x^3) e^{\sec(x^3)} \end{aligned}$$

b) Find $\frac{dy}{dx}$ if $y = \left[\frac{e^x}{e^{2x} + 1} \right]^4$.

Upon our first application of the chain rule we obtain: $\frac{dy}{dx} = 4 \left[\frac{e^x}{e^{2x} + 1} \right]^3 \cdot \frac{d}{dx} \left(\frac{e^x}{e^{2x} + 1} \right)$. We now see that we will have to apply the quotient rule. Specifically,

$$\begin{aligned} \frac{dy}{dx} &= 4 \left[\frac{e^x}{e^{2x} + 1} \right]^3 \cdot \frac{d}{dx} \left(\frac{e^x}{e^{2x} + 1} \right) = 4 \left[\frac{e^x}{e^{2x} + 1} \right]^3 \cdot \frac{(e^{2x} + 1) \frac{d}{dx}(e^x) - e^x \frac{d}{dx}(e^{2x} + 1)}{(e^{2x} + 1)^2} \\ &= \frac{4e^{3x}}{(e^{2x} + 1)^3} \cdot \frac{(e^{2x} + 1)e^x - e^x(e^{2x} \cdot 2)}{(e^{2x} + 1)^2} = \frac{4e^{3x}}{(e^{2x} + 1)^3} \cdot \frac{e^x - e^{3x}}{(e^{2x} + 1)^2} = \frac{4e^{4x}(1 - e^{2x})}{(e^{2x} + 1)^5}. \end{aligned}$$

c) Find $\frac{dy}{dx}$ if $y = (2x+1)^5 (x^3 - x + 1)^4$.

We first note that we will have to apply the product rule, and *then* the chain rule. In particular, we have:

$$\begin{aligned}\frac{dy}{dx} &= (2x+1)^5 \frac{d}{dx} \left[(x^3 - x + 1)^4 \right] + \frac{d}{dx} \left[(2x+1)^5 \right] (x^3 - x + 1)^4 \\ &= (2x+1)^5 \cdot 4(x^3 - x + 1)^3 (3x^2 - 1) + 5(2x+1)^4 (2)(x^3 - x + 1)^4 \\ &= 4(2x+1)^5 (x^3 - x + 1)^3 (3x^2 - 1) + 10(2x+1)^4 (x^3 - x + 1)^4 \\ &= 2(2x+1)^4 (x^3 - x + 1)^3 \left[2(2x+1)(3x^2 - 1) + 5(x^3 - x + 1) \right] \\ &= 2(2x+1)^4 (x^3 - x + 1)^3 [17x^3 + 6x^2 - 9x + 3].\end{aligned}$$

d) Suppose that g is a differentiable function and n is some constant. Find a formula for $\frac{d}{dx} \left[(g(x))^n \right]$.

By the chain rule: $\frac{d}{dx} \left[(g(x))^n \right] = n(g(x))^{n-1} \cdot g'(x)$.

We end this section by using the chain rule to prove an important theorem.

Theorem: Suppose that b is a positive real number constant. Then, $\frac{d}{dx} (b^x) = b^x \ln(b)$.

Proof:

We note that if $y = b^x$, then $y = e^{\ln(b^x)} = e^{x \ln(b)}$. So, by the chain rule

$$\frac{dy}{dx} = e^{x \ln(b)} \cdot \frac{d}{dx} (x \ln(b)) = e^{\ln(b^x)} \ln(b) = b^x \ln(b).$$

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Question: Does the formula for the derivative in the above theorem hold for $y = e^x$?

Question to the Class: Suppose we want to find the first derivative of $y = 2^{2^x}$. Are there any errors in the following two solutions?

Solution 1: By the chain rule and the above theorem: $\frac{dy}{dx} = 2^{2^x} \ln(2) \cdot \frac{d}{dx}(2^x) = 2^{2^x} \ln(2) \cdot 2 = 2^{2^x+1} \ln(2)$

Solution 2: We use some algebra, and we get $y = 2^{2^x} = (2^2)^x = 4^x$. Then, $\frac{dy}{dx} = 4^x \ln(4)$.

Answer: Both solutions are flawless. In fact, we notice that both solutions are equivalent since

$$2^{2^x+1} \ln(2) = 2^{2^x} \cdot 2^1 \cdot \ln(2) = 2^{2^x} \ln(2^2) = 4^x \ln(4).$$

Note: The above question shows that sometimes derivatives can look very different but still be equivalent.